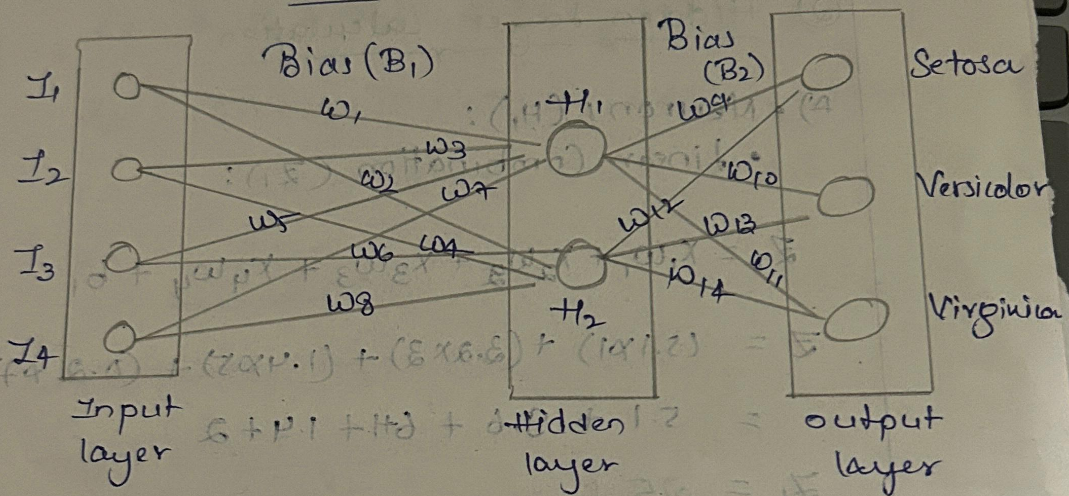
DEEP LEARNING ASSIGNMENT-02

Mathematical Calculation for Single Hidden-Neural-Network-Layer



① Given Parameters:

- Input vectors : $X = [5.1, 3.2, 1.4, 0.2]$
- Hidden weights: (H_1) : $w_1 = 1$, $w_3 = 3$, $w_5 = 5$, $w_7 = 7$
- Hidden Weights $W(H_2)$: $w_2 = 2$, $w_4 = 4$, $w_6 = 6$, $w_8 = 8$
- Output Weights : $w_9 = 1.2$, $w_{10} = 0.5$, $w_{11} = 1.2$
- Output weights : $w_{12} = 0.5$, $w_{13} = 0.2$, $w_{14} = 0.4$

B) Versicolor (Z versicolor):

$$Z_{versicolor} : (\sigma_1 \times \omega_{1,0}) + (\sigma_2 \times \omega_{1,2}) + b_2$$

$$= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1$$

$$= 0.5 + 0.2 + 0.1$$

$$Z_{versicolor} = 0.8$$

c) Virginica (Z virginica):

$$Z_{virginica} : (\sigma_1 \times \omega_{1,1}) + (\sigma_2 \times \omega_{2,4}) + b_2$$

$$= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1$$

$$= 1.2 + 0.4 + 0.1$$

$$Z_{virginica} = 1.7$$

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4 Final Prediction:

- Output Vector : [1.9, 0.8, 1.7]
- Prediction : Setosa [Highest Score : 1.9]

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B): Neuron 2 (H_2):

Linear Combination (Z_2):

$Z_2 = X_1 w_2 + X_2 w_4 + X_3 w_6 + X_4 w_8 + b_1$

$Z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$

$Z_2 = 10.2 + 12.8 + 9.6 + 1.6 + 2$

$Z_2 = 35.0$

Activation (σ_2):

$\sigma_2 = \frac{1}{1+e^{-Z_2}}$

$= \frac{1}{1+e^{-35.0}}$

$\sigma_2 \approx 1.0$

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② Output layer Calculation

A) Setosa (Z setosa):

$Z\ \text{setosa} = (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2$

$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$

$= 1.3 + 0.5 + 0.1$

$\therefore Z\ \text{setosa} = 1.9$

- Biases : $b_1 = 2$ $b_2 = 0.1$

② Hidden Layer Calculation

A) • Neuron 1 (H_1):
  • Linear Combination (Z_1):

$$Z_1 = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b_1$$

$$Z = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2$$

$$= 5.1 + 9.6 + 7 + 1.4 + 2$$

$$Z_1 = 25.1$$

Activation (σ_1):

$$\sigma_1 = \frac{1}{1+e^{-Z_1}}$$

$$\sigma_1 = \frac{1}{1+e^{-25.1}}$$

$$\sigma_1 \approx 1.0$$